

Omar Olivares Urrutia

AI / Machine Learning Engineer — LLM Agents, Evaluation & RAG Systems



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SUMMARY

AI/ML engineer specializing in LLM agents, evaluation, and production RAG systems. Equal parts researcher and builder: benchmarked 19 LLMs on SQL error detection for a peer-reviewed paper (CLEI 2026) and helped launch one of the fastest-growing open-source coding agents. Hands-on across the full ML lifecycle — data collection, fine-tuning, evaluation, and serving at scale — with a focus on AI products people actually use.

SKILLS

AI and Machine Learning

Intelligent Agents · Multi-Agent Systems · Agentic Workflows · LLM Evaluation · LLM-as-a-Judge · Evals · Benchmarking · AI Governance · LLM Observability · RAG Systems · Embeddings · Vector Search · Reranking · Fine-tuning · LoRA · QLoRA · PEFT · Prompt Engineering · Open-Source Development

Frameworks and Tools

PyTorch · Hugging Face · LangChain · OpenAI API · Anthropic API · Weights & Biases · W&B · MLX

Infrastructure and Serving

FastAPI · Docker · Microservices · Model Serving · Inference · Observability · Tracing · PostgreSQL · Vector Databases · Pinecone · ChromaDB · OpenRouter · AWS · GCP

Programming Languages

Python · SQL · TypeScript

WORK EXPERIENCE

Senior AI Engineer at Maxxa

Feb 2026 - Current

📍 Santiago, Chile

Maxxa is a Chilean fintech specializing in credit and surety guarantees. Building the production AI and LLM systems behind its financial products.

- Architected and built the governance, evals, and observability layer for the company's production LLM systems
- Contributing to the internal LLM platform (core and enterprise editions) that powers AI features across the company's financial products
- Helped build the platform's Model Context Protocol (MCP) server, letting AI agents and internal teams plug company data and tools directly into their workflows

AI Engineer at Universidad de Talca

Aug 2025 - Current

Building an AI-powered educational platform for learning SQL, using multi-agent LLM systems to give students the kind of real-time, granular feedback traditional database courses can't.

- Architecting a multi-agent system that gives students real-time feedback on the syntax, semantics, and logic of their SQL queries
- Benchmarking LLM agents — Grok, GPT-4o, Claude 3.5, Gemini, DeepSeek — on educational tasks
- Built an evaluation suite benchmarking 19 LLMs on 300 SQL cases, combining LLM-as-a-Judge scoring with human-annotated ground truth
- Co-authored a peer-reviewed paper on LLM-based SQL error detection, accepted at CLEI 2026 (IEEE, Scopus-indexed)

Staff Developer Advocate at Kilo Code

Feb 2025 - Apr 2025

📍 San Francisco, CA / Amsterdam

Joined the 12-person founding team of Kilo Code, a high-growth open-source coding agent forked from Cline/Roo-Code, from inception through its first scaling phase.

- Helped the founding team launch one of the fastest-growing open-source coding agents
- Built OpenRouter integrations and technical documentation that helped push Kilo Code to the #1 model ranking on OpenRouter
- Wrote integration guides and reference implementations for developer onboarding as the project grew to ~10K GitHub stars
- Shaped technical architecture and strategy working directly with GitLab co-founder Sid Sijbrandij's leadership team

AI Engineer at Emergent Mind

Jan 2024 - Jan 2025

📍 North Carolina, US (Remote)

Led AI development for an arXiv research-discovery platform, building the production RAG systems behind its scientific search and analysis.

- Redesigned the RAG pipeline — chunking, hybrid retrieval, reranking — improving search relevance across 200K+ indexed papers
- Built ingestion and indexing pipelines for daily arXiv submissions and stood up an evaluation harness in Weights & Biases
- Deployed Dockerized FastAPI services with tracing and reliability tests for production serving
- Ran experiments to improve model performance and accuracy, and owned QA for AI-driven features

Content Developer at Neura Pod

Aug 2021 - Dec 2024

📍 San Diego, CA (Remote)

Part-time freelance role held alongside engineering work. Led content for a popular neurotechnology and AI series, translating dense research for a broad audience.

- Turned advanced research into engaging videos, helping grow the channel past 80,000 YouTube subscribers
- Produced multiple viral videos — 200,000+ views each — building public understanding of brain-computer interfaces and AI
- Owned production end to end: research, scripting, and distribution across major social platforms

AI Evaluation Contractor at Centro Nacional de Inteligencia Artificial

Oct 2024 - Nov 2024

📍 Santiago, Chile

Contract engagement running rigorous evaluations of Spanish-language AI models for Latin America.

Software Engineer at clicOH

Jun 2021 - Nov 2021

📍 Remote

Backend engineering at a logistics startup — originally RayoApp, acquired by clicOH.

- Contributed to backend development and systems integration through the post-acquisition transition
- Maintained legacy systems while migrating features to the new architecture

Software Engineering Intern at Suncast

Dec 2019 - Mar 2020

📍 Las Condes, Chile

Summer internship during Computer Science studies, helping rebuild the platform on a scalable microservices architecture.

- Refactored a monolithic application into microservices, improving scalability and maintainability
- Built API endpoints and optimized database queries for a renewable-energy forecasting platform

Information Technology Support Assistant at Universidad de Talca

Dec 2015 - Nov 2018

📍 Curicó, Chile

Provided technical support, resolved network issues, and helped deploy IT infrastructure across the university.

- Diagnosed network issues and rolled out new IT infrastructure alongside the technical team

Music Producer at Ableton

Dec 2007 - Nov 2013

📍 Valencia, Spain

Composed, produced, and engineered tracks for artists.

PROJECTS

WebGrid Eval

Feb 2026 - Current

Skills: LLM Evaluation · Vision-Language Models · Tool Use · Agents · FastAPI · OpenRouter

🔗 https://github.com/ofou/webgrid_eval

Open-source benchmark for LLM vision and tool-use, evaluating models on Neuralink's Webgrid cursor-control task

- Built a benchmark where vision LLMs use screen, mouse_move, and mouse_click tools to drive a cursor, scored in bits-per-second against Neuralink's human baselines
- Ran multi-model evaluations (Gemini, Qwen, and others) through OpenRouter with a FastAPI evaluation server and automated replay generation
- Bridges AI and neurotechnology by scoring LLM agents on the same task Neuralink uses to measure brain-computer-interface performance

Lucas — AI Finance Agent

Nov 2025 - Current

Skills: Anthropic API · Agents · FastAPI

🔗 <https://github.com/platanus-hack/platanus-hack-25-team-29>

Agentic AI personal finance assistant with real-time bank API integration

- Built an AI finance agent that connects to real bank accounts through the Fintoc API
- Shipped in a 36-hour hackathon using the Anthropic Claude API and a FastAPI backend

Textube — YouTube-to-LLM API

Aug 2024 - Current

Skills: FastAPI · LLM Integration · ChatGPT Plugin · Firestore · tiktoken

🔗 <https://github.com/ofou/texttube>

FastAPI service that pipes YouTube transcripts into LLMs, shipped as a ChatGPT plugin — featured on the Hacker News front page

- Featured on the Hacker News front page; shipped as a live ChatGPT plugin and public API
- Built a FastAPI backend with Google Firestore caching and tiktoken token counting, processing 250+ video playlists (1M+ words)
- Drew so much usage that Google rate-limited it — months later, Google shipped similar functionality natively in Gemini

Graham Essays

Jun 2022 - Current

Skills: Python · Web Scraping · Open Source

<https://github.com/ofou/graham-essays>

Automated EPUB generator for Paul Graham's essays — reached #1 on Hacker News, 800+ GitHub stars

- Hit #1 on the Hacker News front page
- Built an automated pipeline that compiles 200+ essays into EPUB format
- Earned 800+ GitHub stars from the open-source community

VOLUNTEER

Software Engineer at Neuromatch Academy

Jun 2021 - Aug 2021

Volunteered as a software engineer with Neuromatch, a global online community of computational neuroscientists focused on inclusive learning, mentorship, and professional development.

EDUCATION

Postgraduate Diploma Generative AI and Large Language Models at Centro Nacional de Inteligencia Artificial (CENIA) Aug 2025 - Jan 2026

Major courses: LLM Fine-tuning (LoRA, QLoRA, PEFT) · Production ML (PyTorch, MLflow, W&B) · Multimodal Models & RAG Systems · Ethical AI & Bias Management

Engineer's degree Computer Science at Universidad Católica del Maule

Feb 2018 - Nov 2023

Major courses:

Thesis: Fine-tuned GPT-3.5 for Chile's PAES math exam — improved accuracy from 78% to 94% (+16 points) · Data Structures & Algorithms, Database Systems, Distributed Systems, AI/ML

Engineer's degree Computer Engineering at Universidad de Talca

Feb 2015 - Nov 2017

AWARDS

2nd Place — Computer Vision Hackathon at Fundación País Digital | EY

Aug 2025

Built a traffic-monitoring system that detects congestion and anomalies in CCTV streams, designed to scale to thousands of cameras.

CERTIFICATES

IELTS Academic - Overall Band Score 7.5 (C1 Level) Cambridge English

Jun 2025

PUBLICATIONS

Assessing the Ability of Large Language Models to Detect and Classify Errors in SQL Queries in CLEI 2026: Latin American Computing Conference (IEEE)

Sept 2026

Full paper at CLEI 2026 (Computer Education track) — IEEE-published, Scopus-indexed, 34% acceptance rate (102 of 297 submissions). The study evaluates how well 19 LLMs detect and classify errors in SQL queries, using LLM-as-a-Judge scoring over 300 test cases against human-annotated ground truth, with direct applications in automated tutoring for database education. To be presented in Mexico City, September 2026.

Optimization of Language Models for Chile's Mathematics Higher Education Admission Test (PAES) in Universidad Católica del Maule

Feb 2024

Fine-tuned GPT-3.5 for Chile's PAES mathematics exam, raising accuracy from 78% to 94% (+16 points) through targeted optimization and OCR/transcription pipelines.

LANGUAGES

Spanish

Native

English

Full Professional (IELTS C1 7.5)

INTERESTS

AI Research

Skills: Interpretability ·
AI alignment & safety ·
Agentic systems ·
Computational neuroscience

Technology

Skills: Neurotechnology ·
Brain-machine interfaces ·
Open source

Art

Skills: Music production ·
Video & content creation ·
Technical writing

Languages

Skills: Learning Mandarin

Travel

Skills: Cultural explorations ·
Adventure travel ·
Culinary experiences

REFERENCES

“ Co-founder, Kilo Code ”

— Matt Mazur

“ Founder, Emergent Mind ”

— Ryan Tanaka